RDB & SQL
SQL: SINGLE TABLE QUERIES

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LEARNING OBJECTIVES

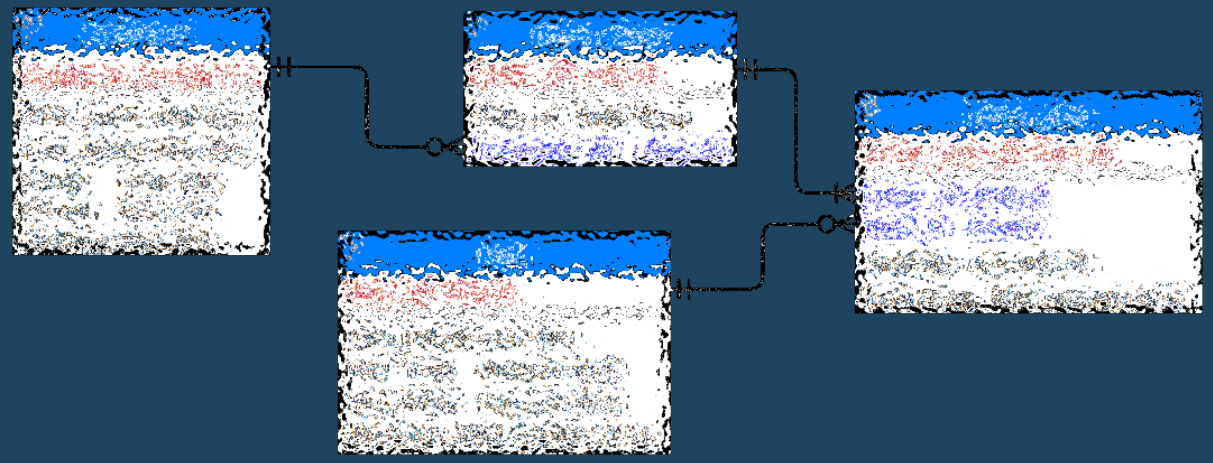
- Learning single table queries with the assistant of AI tools (ChatGPT, Copilot, Gemini, and others)
- Formulate DB query solutions
- Translate query solutions to SQL query statements
- Run the SQL queries to correct solution errors
- Hands-on practices & reflection

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Relational Database & SQL

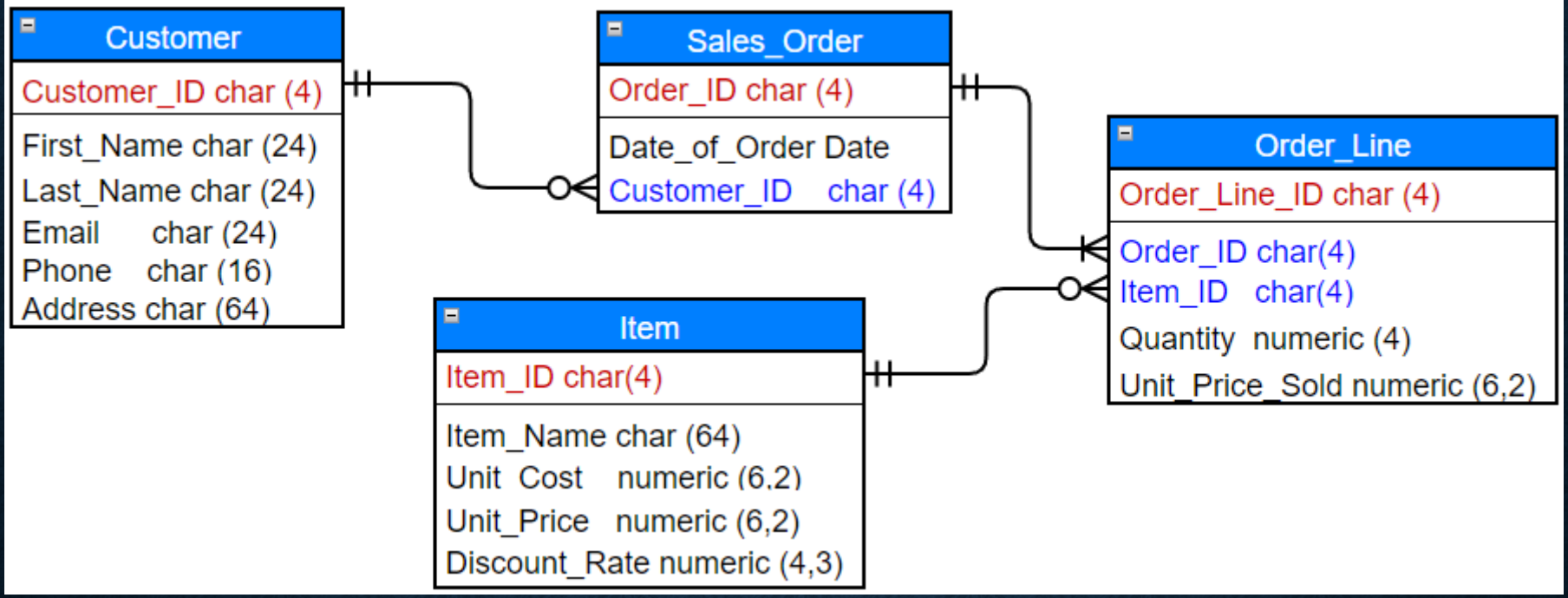
SQL: Single Table Queries

(1) PRE-WORK



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EXERCISE DB: ORDER MANAGEMENT



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WHY'RE DATABASE QUERIES?





Look at the Customer table and think of three business questions you want to know from the table

GenAI

- Ask your AI tool for three business scenarios using the Customer table
- Compare & contrast your questions with the scenarios given by your AI tool
- What is the language used by your AI tool in its answer?

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SQL - THE STANDARD DB LANGUAGE

All relational databases understand the standard programming language, called SQL

- The standard was created by American National Standards Institute (ANSI) in 1986
- The language is called ANSI SQL standard
- It has multiple standard versions, SQL-86, SQL-89, etc.
- SQL-92 is the commonly supported by DB vendors

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HOW TO QUERY THE DATABASE?





Select one of the questions to ask in the Customer table and write the query in SQL

GenAI

- Ask your AI tool to write the SQL query for your question
- Compare & contrast your SQL query with that written by your AI tool.
- Is the SQL query intuitive to understand?

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HOW TO WRITE CORRECT QUERIES IN SQL



GenAI

- Ask your AI tool to explain the syntax of the SQL query in its answer
- Write SQL queries for different questions you want ask on the Customer table
- Enter and run the SQL queries in your Order Management database
- If you get errors, ask your AI tool to help you debug the errors

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HOW TO KNOW IF A QUERY IS CORRECT OR WRONG?



GenAI

- Analyze the query results you got and determine if the results are correct or wrong
- Can you prove if the results are correct?
- Ask your AI tool to prove if the queries it wrote for you are correct
- Are you happy with your AI tool's explanations? Why?

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DB QUERIES SOLVE BUSINESS PROBLEMS

Understand the problem

- What does the user really want?

Formulate a solution

- Understand the DB
- Decide which fields & table should be used to get the answer?

Translate to SQL & test

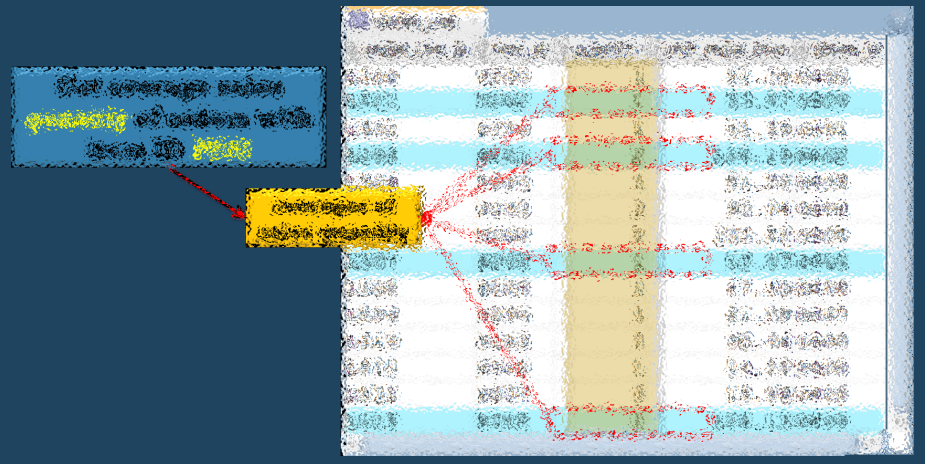
- What SQL commands to use?
- Is the answer correct?

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Relational Database & SQL

SQL: Single Table Queries

(2) FORMULATE SOLUTIONS



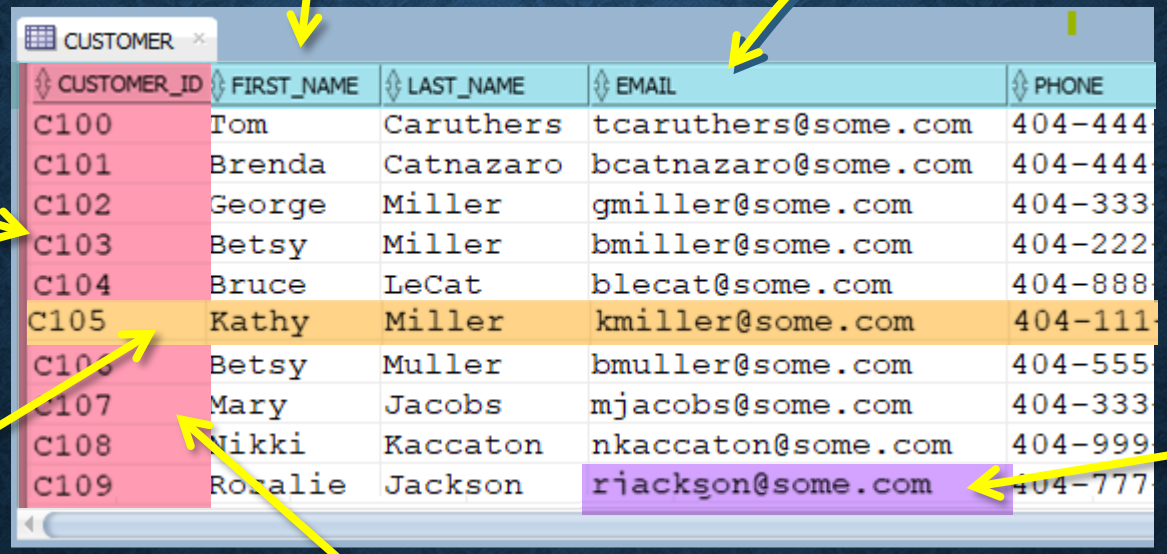
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TABLE (RELATION) REVISITED

Table
(*relation*)

Table name
(*relation name*)

Field names & Types
(*Attribute names & domains– **meta-data***)



Primary key
Unique identifier for each tuple
(unique & minimal)

Field value
(*attribute value*)

*Every field value can have only a single value (**atomic**)*

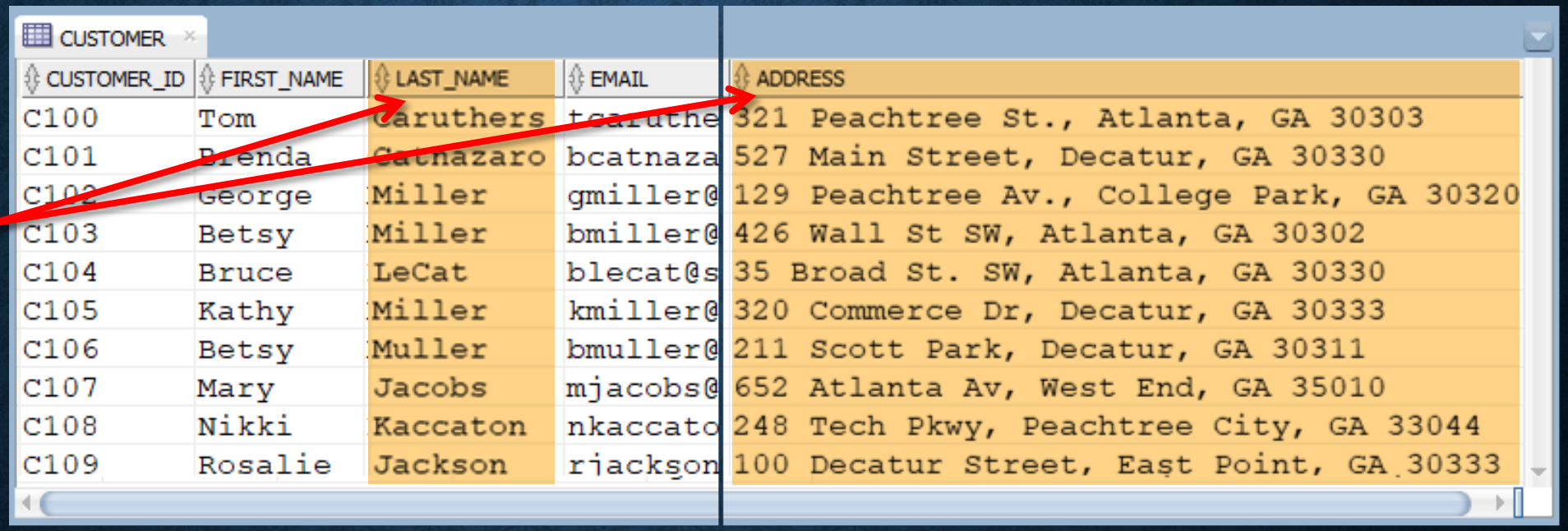
Rows
(records/tuples)
Each row represents an object In the real world

Columns/fields
(*attributes*)

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SELECTING DESIRED DATA FIELDS (COLUMNS)

List customer's last name and addresses



CUSTOMER

CUSTOMER_ID	FIRST_NAME	LAST_NAME	EMAIL	ADDRESS
C100	Tom	aruthers	tcaruthe	321 Peachtree St., Atlanta, GA 30303
C101	Brenda	atnazaro	bcatnaza	527 Main Street, Decatur, GA 30330
C102	George	Miller	gmiller@	129 Peachtree Av., College Park, GA 30320
C103	Betsy	Miller	bmiller@	426 Wall St SW, Atlanta, GA 30302
C104	Bruce	LeCat	blecat@s	35 Broad St. SW, Atlanta, GA 30330
C105	Kathy	Miller	kmiller@	320 Commerce Dr, Decatur, GA 30333
C106	Betsy	Muller	bmuller@	211 Scott Park, Decatur, GA 30311
C107	Mary	Jacobs	mjacobs@	652 Atlanta Av, West End, GA 35010
C108	Nikki	Kaccaton	nkaccato	248 Tech Pkwy, Peachtree City, GA 33044
C109	Rosalie	Jackson	rjackson	100 Decatur Street, East Point, GA 30333

GenAI Work with your AI tool to formulate a solution to the given problem and explain the solution

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SELECTING DESIRED DATA RECORDS (ROWS)

Who are customers lived in downtown Atlanta?

CUSTOMER

CUSTOMER_ID	FIRST_NAME	LAST_NAME	EMAIL	ADDRESS
C100	Tom	Caruthers	tcaruthe	321 Peachtree St., Atlanta, GA 30303
C101	Brenda	Catnazaro	bcatnaza	527 Main Street, Decatur, GA 30330
C102	George	Miller	gmiller@	129 Peachtree Av., College Park, GA 30320
C103	Betsy	Miller	bmiller@	426 Wall St SW, Atlanta, GA 30302
C104	Bruce	LeCat	blecat@s	35 Broad St. SW, Atlanta, GA 30330
C105	Kathy	Miller	kmiller@	320 Commerce Dr, Decatur, GA 30333
C106	Betsy	Muller	bmuller@	211 Scott Park, Decatur, GA 30311
C107	Mary	Jacobs	mjacobs@	652 Atlanta Av, West End, GA 35010
C108	Nikki	Kaccaton	nkaccato	248 Tech Pkwy, Peachtree City, GA 33044
C109	Rosalie	Jackson	rjackson	100 Decatur Street, East Point, GA 30333

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SELECTING DESIRED FIELDS OF SPECIFIC RECORDS

Get the ***Oder IDs*** of sales orders which were placed in **October** by the customer whose customer ID is **C101**

SALES_ORDER

ORDER_ID	DATE_OF_ORDER	CUSTOMER_ID
O400	05-OCT-18	C101
O401	01-NOV-18	C101
O402	10-SEP-18	C103
O403	01-OCT-18	C103
O404	15-NOV-18	C103
O405	20-OCT-18	C102
O406	03-DEC-18	C101
O407	02-DEC-18	C103
O408	15-SEP-18	C108
O409	25-NOV-18	C106

GenAI Work with your AI tool to write the SQL query for the given problem

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GET STATISTICAL FACTS ABOUT ALL DATA

Get average sales *quantity* of *all* orders

Computed the average of all sales quantity

ORDER_LINE

ORDER_LINE_ID	ITEM_ID	QUANTITY	UNIT_PRICE_SOLD	ORDER_ID
L500	P200	1	51.29	O400
L501	P202	1	53.07	O400
L502	P200	1	51.29	O401
L503	P202	2	106.13	O402
L504	P200	1	51.29	O403
		2	92.04	O404
		2	102.58	O405
		1	53.07	O405
L508	P206	1	47.49	O406
L509	P200	1	51.29	O407
L510	P203	2	92.04	O408
L511	P205	1	14.28	O408

GenAI Work with your AI tool to formulate a solution to the given problem and explain its solution

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CALCULATING STATISTICAL FACTS ILLUSTRATED

Get average sales *quantity* of orders with item ID *P202*

averages of sales quantity

ORDER_LINE

ORDER_LINE_ID	ITEM_ID	QUANTITY	UNIT_PRICE_SOLD	ORDER_ID
L500	P200	1	51.29	O400
L501	P202	1	53.07	O400
L502	P200	1	51.29	O401
L503	P202	2	106.13	O402
L504	P200	1	51.29	O403
	P203	2	92.04	O404
	P200	2	102.58	O405
L507	P202	1	53.07	O405
L508	P206	1	47.49	O406
L509	P200	1	51.29	O407
L510	P203	2	92.04	O408
L511	P205	1	14.28	O408

GenAI Work with your AI tool to formulate a solution to the given problem and explain its solution

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USING DATA PATTERNS TO SELECT ROWS

Find the customer whose last name is *"Miller"* and first name ends with the letter *'e'*

CUSTOMER

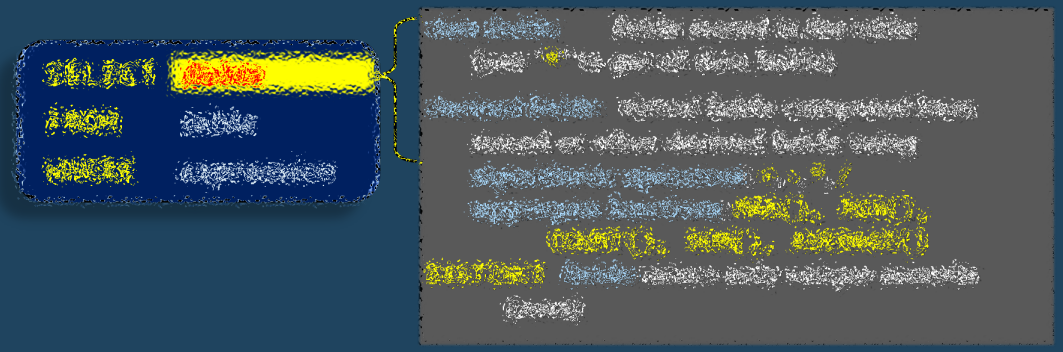
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C101	Brenda	Catnazaro	bcatnaza	527 Main Street, Decatur, GA 30330
C102	George	Miller	gmiller@	129 Peachtree Av., College Park, GA 30320
C103	Betsy	Miller	bmiller@	426 Wall St SW, Atlanta, GA 30302
C104	Bruce	LeCat	blecat@s	35 Broad St. SW, Atlanta, GA 30330
C105	Kathy	Miller	kmiller@	320 Commerce Dr, Decatur, GA 30333
C106	Betsy	Muller	bmuller@	211 Scott Park, Decatur, GA 30311
C107	Mary	Jacobs	mjacobs@	652 Atlanta Av, West End, GA 35010
C108	Nikki	Kaccaton	nkaccato	248 Tech Pkwy, Peachtree City, GA 33044

GenAI Work with your AI tool to formulate a solution to the given problem and explain its solution

Relational Database & SQL

SQL: Single Table Queries

(3) SQL QUERY SYNTAX



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SQL – FIELD SELECTION

```
SELECT   fields
FROM     table
WHERE    expression
```

data fields: fields stored in the table
(use "*" to get all data fields)

derived fields: virtual fields computed from stored or other derived fields using
algorithmic operators: +, -, *, /
aggregate functions: MAX(), MIN(), COUNT(), SUM(), AVERAGE()

DISTINCT *fields*: return only unique records (rows)

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SQL – TABLE SELECTION

```
SELECT fields
FROM   table
WHERE  expression
ORDER BY fields
```

A single table in this learning unit

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SQL – ROW SELECTION

```
SELECT fields
FROM   table
WHERE  Boolean Expression
ORDER BY fields
```

field [=, <>, >, >=, <, <=] *field*

field [**NOT**] **IN** (*value1*, *value2*, ...)

test if the *field* contains one of those *values* (doesn't contain any of those *values*)

field **BETWEEN** *value1* **AND** *value2*

test if the value of the *field* falls within the range delimited by *value1* and *value2*

field **LIKE** *pattern-matching*

_ match any single character

% match any one or more characters

[**NOT**] *expression*

[**(**] *expression* [**AND**, **OR**] *expression* [**)**]

A complex expression

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SQL – SORTING QUERY RESULTS

```
SELECT fields
FROM   table
WHERE  expression
ORDER BY fields
```

field [**ASC**]
sort the query result by the *field* in ascending order, the default sorting if ASC is absent

field **DESC**
sort the result by the *field* in descending order

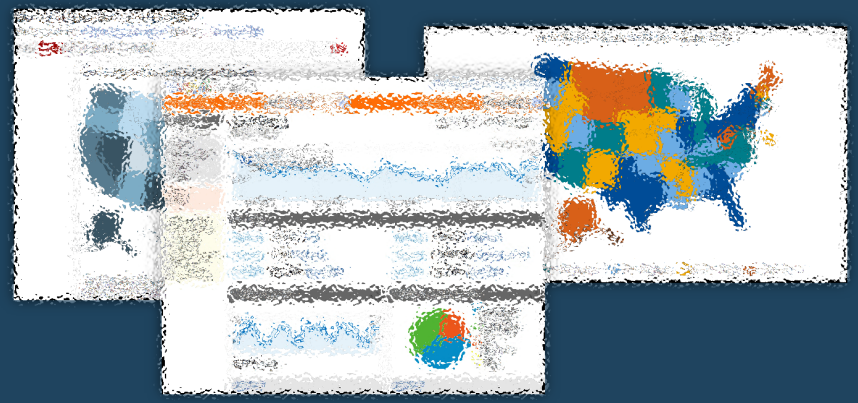
field 1, field 2, … field n
sort the result by *field 1* first, *field 2* next, and so on

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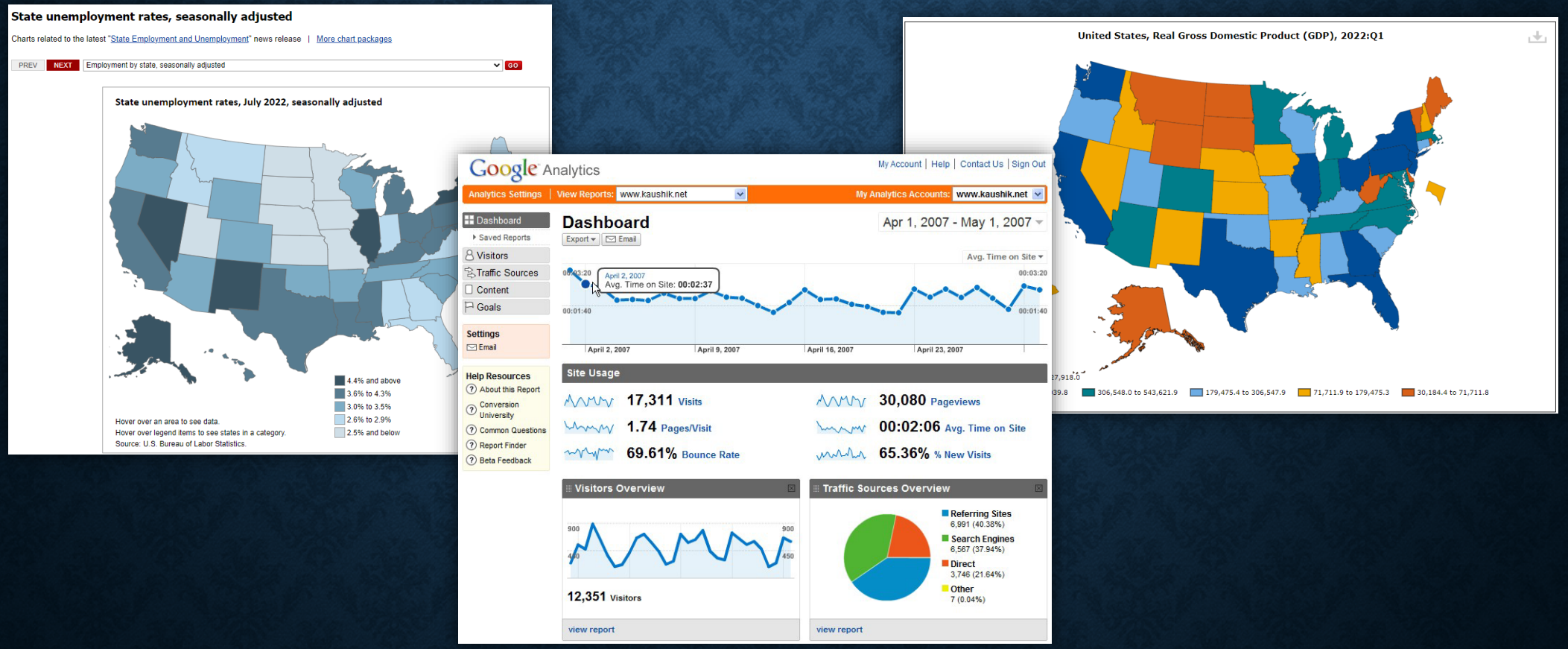
SQL: Single Table Queries

(4) GROUP AGGREGATE DATA



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GROUP AGGREGATE DATA EXAMPLES



Calculate the total sales *quantity* for *each item*

Sum up the sales qualities of item P200

Repeat the step for each of the remaining items, P202, P203, …

ORDER_LINE

ORDER_LINE_ID	ITEM_ID	QUANTITY	UNIT_PRICE_SOLD	ORDER_ID
L500	P200	1	51.29	O400
L501	P202	1	53.07	O400
L502	P200	1	51.29	O401
L503	P202	2	106.13	O402
L504	P200	1	51.29	O403
L505	P203	2	92.04	O404
L506	P200	2	102.58	O405
L507	P202	1	53.07	O405
L508	P206	1	47.49	O406
L509	P200	1	51.29	O407
L510	P203	2	92.04	O408
L511	P205	1	14.28	O408
L512	P207	3	15.86	O408
L513	P202	2	106.13	O409

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Calculate the total sales *quantity* sold for *each item*

A better approach is to group all sales by Item ID first

ORDER_LINE

ORDER_LINE_ID	ITEM_ID	QUANTITY	UNIT_PRICE_SOLD	ORDER_ID
L500	P200	1	51.29	O400
L506	P200	2	102.58	O405
L509	P200	1	51.29	O407
L504	P200	1	51.29	O403
L502	P200	1	51.29	O401
L513	P202	2	106.13	O409
L507	P202	1	53.07	O405
L501	P202	1	53.07	O400
L503	P202	2	106.13	O402
L505	P203	2	92.04	O404
L510	P203	2	92.04	O408
L511	P205	1	14.28	O408
L508	P206	1	47.49	O406
L512	P207	3	15.86	O408

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Calculate the total sales *quantity* for *each item*

… and then calculate the sum of sales quantity in each item group

ORDER_LINE

ORDER_LINE_ID	ITEM_ID	QUANTITY	UNIT_PRICE_SOLD	ORDER_ID
L500	P200	1	51.29	O400
L506	P200	2	102.58	O405
L509	P200	1	51.29	O407
L504	P200	1	51.29	O403
L502	P200	1	51.29	O401
L513	P202	2	106.13	O409
L507	P202	1	53.07	O405
L501	P202	1	53.07	O400
L503	P202	2	106.13	O402
L505	P203	2	92.04	O404
L510	P203	2	92.04	O408
L511	P205	1	14.28	O408
L508	P206	1	47.49	O406

SUM()

SUM()

SUM()

SUM()

SUM()

GenAI Work with your AI tool to write the SQL query for the given problem

Calculate the total sales *quantity* for *each item*

… and then calculate the sum of sales quantity in each item group

```
SELECT  SUM(Quantity)
FROM    Order_Line
GROUP BY Item_ID;
```

ORDER_LINE

ORDER_LINE_ID	ITEM_ID	QUANTITY	UNIT_PRICE_SOLD	ORDER_ID
L500	P200	1	51.29	O400
L506	P200	2	102.58	O405
L509	P200	1	51.29	O407
L504	P200	1	51.29	O403
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L503	P202	2	106.13	O402
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L510	P203	2	92.04	O408
L511	P205	1	14.28	O408
L508	P206	1	47.49	O406
L512	P207	3	15.86	O408

SUM()

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EXPLORE GROUP AGGREGATE DATA QUERIES



GenAI

- Ask your AI tool for a SQL query for the same problem in the previous slide
- Compare & contrast AI's answer with that given in the previous slide.
- Modify the problem to show both the total sales quantity and total sales amount for each item
- Test the SQL query in your database and work with your AI tool to fix errors if any

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Interested only those items whose total quantity sold is greater than or equal to 4 units

Select only those item groups whose total quantity sold ≥ 4

```
SELECT  SUM(Quantity)
FROM        Order_Line
GROUP BY Item_ID
HAVING SUM(Quantity) >= 4;
```

ORDER_LINE

ORDER_LINE_ID	ITEM_ID	QUANTITY	UNIT_PRICE_SOLD	ORDER_ID
L500	P200	1	51.29	O400
L506	P200	2	102.58	O405
L509	P200	1	51.29	O407
L504	P200	1	51.29	O403
L502	P200	1	51.29	O401
L513	P202	2	106.13	O409
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L511	P205	1	14.28	O408
L508	P206	1	47.49	O406
L512	P207	3	15.86	O408

6

6

4



PRACTICE GROUP AGGREGATE DATA



GenAI

- Ask your AI tool to write a SQL query for the same problem in previous slide
- Compare & contrast its answer with the solution described in previous slide
- Enter and run those SQL statements in your database to get the query results
- Use your AI tool to help you debug errors encountered

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GROUP SELECTION CHALLENGE



GenAI

- Work with your AI tool to write a SQL query to retrieve only those items whose total quantity sold ≥ 4 or total sales amount ≥ $400
- Test the SQL query in your database with the help of your AI tool

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UNDERSTAND GROUP SELECTION





Are the WHERE and HAVING clauses interchangeable?

GenAI

- Describe your answer to your AI tool and ask it to assess whether your answer is correct or not
- If you still have difficulties to understand the difference, please ask your AI tool for explanations and examples.

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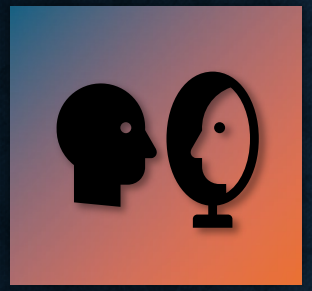
Relational Database & SQL

SQL: Single Table Queries

TAKEAWAYS...



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TAKEAWAYS…

- Good understanding of problems is the key to write queries
- The basis of SQL query techniques are
  - Selecting specific columns in the SELECT clause
  - Selecting specific rows in the WHERE clause
  - Grouping rows using the GROUP-BY clause
  - Selecting specific groups using the HAVING clause
- AI tools can assist in learning new concepts and help comprehend the concepts
- Practice makes perfect

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CONTINUE LEARNING & PRACTICE



Following are very helpful online resources:

- W3schools.com on SQL Tutorial
- Khan Academy on Intro to SQL
- Sololearn course on Introduction to SQL

Those are only a few selected good courses on SQL. You should explore others widely available

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